ALLEGRA WONG

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EDUCATION

The Cooper Union for the Advancement of Science & Art | New York, NY Expected May 2028
Bachelor of Mechanical Engineering | GPA: 3.43
Relevant Coursework: Dynamics, Materials Science, Statics, Design & Prototyping, Thermodynamics, Dynamic Systems Modeling
Awards: Half-Tuition Scholarship, Samuel & Sally Gilman Scholarship Fund, Cooper Hawks Grant, C.V. Starr Scholarship

WORK EXPERIENCE

Product Design Intern March 2026 – Present
The Loisaida Center | New York, NY,
- Design & build air-monitoring devices to support environmental health initiatives in senior housing.
- Improve ease of electronics maintenance, reduce assembly time within a four-person team, enabling rollout of 10 units across Loisaida, with expansion plan of 100 units for a senior housing installation.

Research Assistant Sept 2025 – Present
Bio-Inspired Design Lab | New York, NY
- Fabricate & design a low-cost, open-source, aerial micro robot to model flight biomimicry within a three-person team to optimize design reproducibility & accessibility.
- Model & assemble transmission systems in Fusion 360 to integrate encoders for motor feedback.
- Document prototypes, performance analysis, design decisions, & handover process for future development.

Engineering Intern June – August 2025
RCCM Solutions, LCC | Syosset, NY
- Prepared mechanical equipment layout drawings using AutoCAD LT, system diagrams (chilled water, hot water, airflow), & performed ductwork/piping sizing & pressure-drop calculations.
- Reviewed NYC Building Code & integrated code compliance into mechanical design documentation & data analysis workflows.

Lab Tech Assistant May – July 2025
The Cooper Union | New York, NY
- Managed inventory tracking & organized lab materials to improve workflow efficiency & safety.
- Redesigned equipment layout to improve accessibility & ergonomics; coordinated space setup for academic use.

PROJECTS

Ross Robotics Sept 2025 – Present
MILILAB | New York, NY
- Design throwable, low-cost robot, capable of using camera vision to map structurally instable environments.
- 3D model spherical robot prototype enclosure & structural frame in Fusion 360 & Onshape.

LEADERSHIP ACTIVITIES

Society of Women's Engineers May 2025 – Present
Social Media Manager | New York, NY
- Develop & manage digital & physical content to promote events & increase member engagement

SKILLS

Technical: Onshape, Fusion360, LaTeX, AutoCAD, PrusaSlicer, 3D printing, Microsoft Office (Word, Excel, PowerPoint, Outlook), Canva, Google Workspace